

Learning & Development Analytics Platform

Replacing the L&D Main Reports workbook with an automated analytics application sourced directly from the CRM Learning Programs module and the HRIS.

Requesting department	Learning & Development
Delivery team	Internal Engineering
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Source documents	L&D Main Reports.xlsx; L&D Report Steps.docx; CRM Learning Programs module

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Revision history

Version	Date	Author	Change
1.0	30 Aug 2026		Initial draft for stakeholder review
1.1	30 Aug 2026		P-03 restated: the defect is a hidden evaluation filter, not an averaging error. P-07 overlap corrected (38 rows / 12 people, not 52). P-12 added (# is not a session key). Section 14 reconciliation populated with measured quality-score deltas. Q-11 and Q-12 added.

Approvals

Role	Name	Signature	Date
L&D Director			
HR Leadership			
Engineering Lead			
CRM System Owner			

1. Executive summary

The L&D department produces its core management reporting in a single Excel workbook assembled by hand every cycle. The process takes an estimated 8 to 12 hours per cycle, and the resulting numbers contain defects that can be demonstrated from the file itself.

This document specifies a replacement: a web application that reads Learning Programs, sessions, enrollments, attendance and feedback from the CRM via REST API, joins them to live HRIS headcount data, computes every published metric from a single audited definition, and presents them in an interactive dashboard with automated, exportable reporting.

The build is a custom full-stack application – Python/FastAPI services over PostgreSQL with a React front end – chosen so the platform can integrate directly with the CRM ecosystem and evolve into a broader L&D tool rather than remaining a reporting appliance.

8 to 12 hours	25 of 55	3	19%
Manual effort per reporting cycle that	Programs visible in the workbook. It covers	Proven broken formulas in	Survey response rate underlying all quality

the platform removes.	under half of actual CRM activity.	currently published KPIs.	scores. Currently measured by nobody.
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2. Current state

All L&D management reporting lives in `L&D Main Reports.xlsx` – twelve worksheets combining raw inputs, hand-built pivot tables and a single-row dashboard. A companion SOP, `L&D Report Steps.docx`, documents a seven-step manual assembly process.

2.1 The existing process

1. Attendance sheets are extracted from the CRM, one per training day.
2. They are pasted into the Attendance Prep tab.
3. Seven attributes are then typed in by hand per block: enrolled count, Public Calendar versus Customised (twice, at participation and topic level), the owning department for customised programs, learning hours, internal versus external, and sector via `VLOOKUP` against the headcount sheet.
4. The enriched block is copied into Attendance Master Sheet.
5. Fourteen pivot tables in Program Pivots are refreshed.
6. The survey export is pasted into Evaluation Master Sheet, where ten helper columns of `IF()` formulas derive NPS bands and quality flags.
7. The DASHBOARD tab reads nine KPIs out of those pivots through `GETPIVOTDATA` strings.

2.2 Current data footprint

Dataset	Records	Coverage	Source
Attendance records	415	25 programs, 128 learners, 39 program-days, Feb to Aug 2026. Session identity is unreliable – see P-12.	CRM export, hand-enriched
Evaluation responses	77	11 programs, 5 trainer name variants	Survey export
Employee headcount	212	Duplicated into two sheets to work around <code>VLOOKUP</code>	HRIS export
CRM Learning Programs	55	Only 25 reach the workbook	CRM (system of record)

3. Problem statement

Every finding below was verified against the delivered workbook. These are not hypotheticals – they are defects currently present in reported numbers.

ID	Finding	Evidence	Impact
P-01	Headcount denominator is hardcoded and wrong.	Dashboard divides by 192; Program Pivots!C1 also holds 192. The HR sheet contains 212 rows and is named "Head Count 212".	Participation rate overstated by approximately 10%. Reported 66.7%; actual 60.4%.
P-02	Pivots key on program title, not ID.	CRM holds Hard Talks as both #87 and #81, and From Individual Actions To Company Impact as #78 and #77.	Separate deliveries silently merge into one program. Program counts and per-program averages are understated.
P-03	NPS and three quality scores are published over a silently filtered subset.	The Evaluation Pivots source carries an undisclosed Completion Time filter. It drops 22 of 77 responses – The Influence Map (Projects) (18) and inside out (4). The remaining 55 give 51 promoters and 0 detractors: 51/55 = 0.9272727, the dashboard value to seven decimals.	29% of feedback is excluded, including the two largest programs. NPS reads 92.7% instead of 84.4%; Logistics 96.4% instead of 92.2%; Knowledge 100% instead of 97.4%. Every quality score is flattered.
P-04	Trainer names are not conformed.	Ahmed Elshiaty (30 responses) and Ahmed ElShiaty (11) are treated as two people.	One trainer's evaluation record is split across two rows; neither reflects his real performance.
P-05	Categorical values carry trailing whitespace.	Internal versus Internal (8 rows); Projects (28) versus Projects (41).	The Projects sector splits into two pivot rows. Sector totals are wrong on both.
P-06	Three grains are stored in one flat table.	Program Duration/hrs is populated on 39 of 415 rows, No. of sessions on 33, Enrollments on 26 – only the first row of each block.	Sums are correct only by convention. A single accidental fill silently doubles reported hours, with no validation to catch it.

P-07	Learners cannot always be joined to HR.	38 rows carry Employee Code = 'N/A'; the 14 rows with Email = 'New Hire' are a subset of those 38, not additional. The 38 rows represent 12 distinct people . The sector pivot has an #N/A bucket.	Those attendees disappear from every sector and department breakdown without warning. Do not report 52 – the two figures overlap completely.
P-08	Under half of training activity is reported.	CRM lists 55 Learning Programs; the workbook contains 25.	Leadership sees a materially incomplete picture of L&D output.
P-09	Top Learner and LinkedIn tracking are fully manual.	Hand-typed name lists, a hardcoded top-four, and =B95 pointing at a count cell.	Not reproducible, not auditable, and quietly stale between updates.
P-10	An unresolved external workbook link exists.	Top Learner!E3 references [1]Sheet1 – a file not present alongside the workbook.	The report cannot be rebuilt from the delivered file alone.
P-12	The # column is not a usable session key.	117 distinct values across 415 rows, ranging 1 to 166. 99 values appear exactly once (row-counter behaviour) while 18 repeat , one covering 46 rows (session-ID behaviour). Separately: 39 distinct dates, 39 distinct program-and-date pairs, and a No. of sessions column summing to 40.	Backfill blocker. COUNT(DISTINCT session_key) over this column is meaningless, and a UNIQUE(employee, session) constraint keyed on it rejects 90 rows. Session identity must come from the CRM, not the workbook.
P-11	No enrollment funnel or response-rate measure exists.	77 evaluations against 415 attendance records.	A 19% survey response rate is invisible, so all quality scores rest on an unexamined sample.

4. Objectives and success criteria

ID	Objective	Measure of success
0-01	Eliminate manual report assembly	Cycle effort falls from 8 to 12 hours to under 30 minutes, limited to enrichment review and exception clearing.
0-02	Report on all training activity	100% of CRM Learning Programs appear in the dashboard, versus 45% today.
0-03	Make every published number auditable	Each KPI traces to one named definition, and drills through to the source records that produced it.
0-04	Correct the defective metrics	P-01 through P-05 resolved, with a signed-off reconciliation of old versus new values.
0-05	Answer questions Excel cannot	Coverage gaps, enrollment funnel and trainer scorecards available as standing views.
0-06	Automate distribution	Scheduled monthly report delivered by email with no human step.
0-07	Surface data problems instead of hiding them	Unresolvable records enter an exception queue; they are never silently dropped.

5. Scope

5.1 In scope

- CRM integration: programs, sessions, enrollments, attendance, feedback
- HRIS integration: employee master with history
- Dimensional data model and single metric layer
- Enrichment administration for the two fields the CRM does not supply
- Interactive dashboard with filters and full drill-down
- Coverage, funnel and scorecard analyses
- Excel, CSV and PDF export
- Scheduled report generation and email delivery
- Data-quality exception console
- Historical backfill of February to August 2026 from the existing workbook

5.2 Out of scope

- Any write-back to the CRM. The platform is read-only against it.
- Replacing CRM data capture, enrollment or QR attendance
- Building or hosting the feedback survey
- Access for sector heads or executives, and the row-level security it would require
- Training budget, cost and ROI analysis. No cost data exists in either source system.
- Plan-versus-actual against an annual training plan. No plan data exists yet.
- LinkedIn Learning integration. Deferred pending an available data feed.
- Skills, competency or certification tracking

DEFERRED, NOT DROPPED

Plan-versus-actual and LinkedIn Learning are both represented in the current workbook, as an empty Planned column and a manual Linked In no sheet. Both are excluded here only because the required source data does not exist. The data model must not preclude them. See section 16.

6. Users

The platform serves one primary user group. This is a deliberate narrowing that removes the need for row-level security in version 1 and allows a dense, operational interface rather than an executive-polished one.

Role	Needs	Frequency
L&D Specialist	Clear the exception queue, complete program enrichment, verify a cycle is reportable, pull ad-hoc cuts.	Daily
L&D Manager	Program and trainer performance, coverage gaps, no-show patterns; produce the monthly report.	Weekly
L&D Director	Departmental performance at a glance; export a report pack for HR leadership.	Monthly

All three share one permission set in version 1. Reports produced for HR leadership and executives are distributed as exported files, not as platform logins.

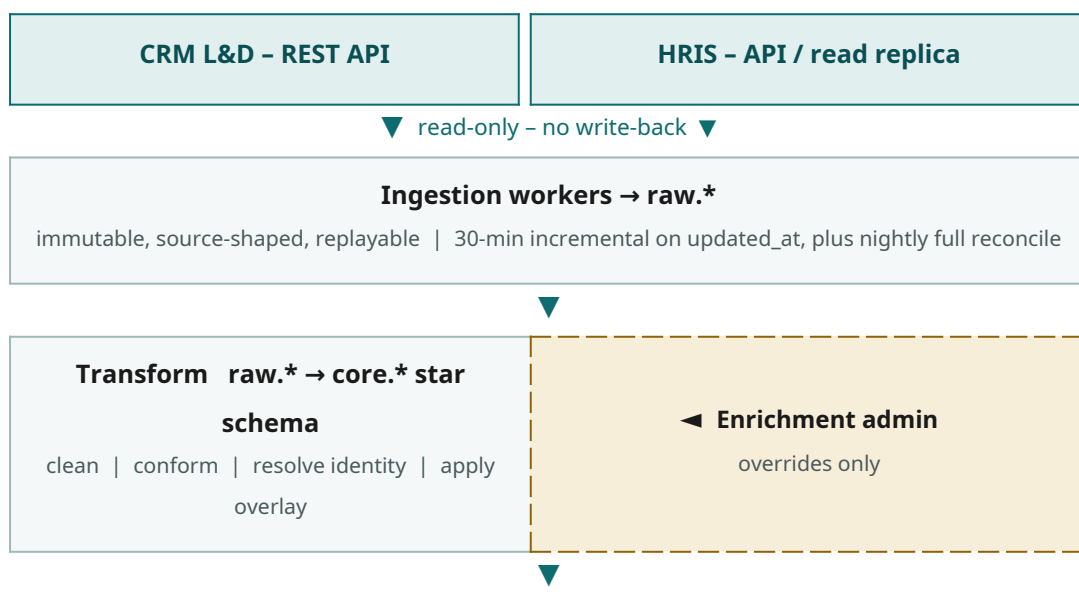
7. Source systems and integration

The CRM Learning Programs module already holds most of what L&D currently types by hand. Confirming this materially reduces build scope: the anticipated program-catalogue administration tool collapses to a two-field mapping screen.

7.1 Field availability

Workbook column (hand-entered today)	Source	Status
Participation PC/Cust	CRM – Program Target (Public / Department)	Available
Internal / External	CRM – Program Type	Available
No. of sessions	CRM – Sessions Count	Available
Program Duration / hrs	CRM – derived from session start and end times	Derivable
Enrollments	CRM – Enrollments tab	Available
(not in workbook)	CRM – Capacity, Status, Fully Booked, Track Title	New capability
Customised (Dept)	Not in CRM. Target says "Department" but not which one.	Enrichment
Trainer Name	Not confirmed present in CRM	Enrichment
Sector	HRIS – joined on employee identity	Join

7.2 Integration architecture



Metric layer – one definition per KPI

shared by the API, the exports and the test suite



FastAPI /v1/ → React dashboard + scheduled report jobs

7.3 Rationale for three key decisions

The raw layer is immutable. CRM responses are stored exactly as received, before cleaning. When a number looks wrong, this is what proves whether it arrived wrong or the platform broke it. It also allows the full history to be replayed after a transform fix without re-querying the CRM. At this data volume the storage cost is negligible.

Enrichment is an overlay, never an edit. L&D's values live in a separate table keyed to CRM program ID and are applied during transform. Synced data is never mutated. If the CRM later adds a trainer field, the source is switched and the overrides retire with no migration.

Sync is both incremental and full. Incremental pulls keep the dashboard current; a nightly full pull catches deletions and silent edits that incremental sync misses. At 55 programs a full pull costs seconds.

8. Data model

The workbook's core structural fault is that one flat table carries three different grains (P-06). Separating them is the foundation of the fix.

8.1 Dimensions

Table	Key	Notes
dim_employee	employee_code	Slowly Changing Dimension Type 2. Participation rate requires headcount as of the period, not today's. Without versioning, February's rate silently changes every time someone joins or leaves – the same instability as the hardcoded 192. At 212 rows, versioning is free.
dim_program	crm_program_id	Keyed on CRM ID, never title. This is the direct fix for P-02.
dim_session	crm_session_id	Child of program. duration_hours derived from start and end times.
dim_trainer	canonical_name	Carries an alias table so name variants resolve to one key (P-04).
dim_date	date	Calendar with month, quarter and fiscal attributes.

8.2 Facts: three tables, three grains

Table	Grain	Rows today	Measures
fact_enrollment	one employee x one program	approx. 409	enrolled_at, status
fact_attendance	one employee x one session	415	attended, learning_hours
fact_evaluation	one response x one program	77	q1 to q5, comment, nps_band

FINDING: TWO HOUR METRICS, ONE NAME

The workbook already tracks two distinct hour measures and names neither. Sum of Program Duration/hrs = **130.5** is catalogue hours: how much distinct training was built and delivered. Sum of Learning Hrs = **1,386** is learner-hours: hours delivered multiplied by the people who received them. They differ by an order of magnitude and both are correct, for different questions. The platform will name them **Training Hours Delivered** and **Learner Hours** so they are never again conflated.

8.3 Supporting tables

Table	Purpose
enrichment_program_override	Customised department and trainer, per CRM program ID, with author and timestamp. The only human-authored data in the system.
identity_resolution	Matches attendance to employees by employee code, then email, then normalised name. Anything unmatched is quarantined rather than dropped (P-07).
sync_run	Audit trail of every sync: source, window, records processed, duration, outcome.
dq_exception	Open data-quality issues awaiting human resolution.

9. Metric definitions

Every metric is defined exactly once, in the metric layer, and consumed identically by the API, the exports and the test suite. This replaces nine KPIs currently defined inside nine unauditable GETPIVOTDATA strings.

Metric	Definition	Versus workbook
Total Programs	COUNT(DISTINCT program_key) WHERE status = Completed	Changed. Was distinct count of title (P-02).
Training Days	COUNT(DISTINCT session_key)	Same intent. No longer a manual sum.
Total Participants	COUNT(DISTINCT employee_key) in fact_attendance	Unchanged.
Training Hours Delivered	SUM(dim_session.duration_hours)	Same intent. Derived from times, not typed.
Learner Hours	SUM(fact_attendance.learning_hours)	Unchanged.
Participation Rate	participants / active headcount as of period end	Corrected. Was divided by hardcoded 192 (P-01).
NPS	(SUM(promoters) - SUM(detractors)) / SUM(responses), aggregated over the group	Corrected. Formula unchanged; scope corrected – computed over all responses, not the filtered 55 (P-03). 92.7% to 84.4%.
Facilitator Performance	COUNT(q4 >= 4) / COUNT(responses)	Unchanged. Scope corrected (P-03); value happens to be identical at 100.0%.
Knowledge Relevance	COUNT(q1 >= 4) / COUNT(responses)	Corrected. Formula unchanged; scope corrected (P-03). 100.0% to 97.4%.
Activity Effectiveness	COUNT(q2 >= 4) / COUNT(responses)	Corrected. Formula unchanged; scope corrected (P-03). 98.2% to 97.4%.
Logistics Effectiveness	COUNT(q3 >= 4) / COUNT(responses)	Corrected. Formula unchanged; scope corrected (P-03). 96.4% to 92.2%.
Internal %	programs WHERE type = Internal / all programs	Same intent. From CRM, not hand-typed.
Public vs Customised split	grouped by dim_program.target	Same intent. From CRM, not hand-typed.
No-show Rate	(enrolled - attended at least one	New.

	session) / enrolled	
Fill Rate	enrollments / dim_program.capacity	New. Capacity exists only in CRM.
Survey Response Rate	COUNT(evaluations) / COUNT(DISTINCT attendees)	New. Currently 19%, untracked (P-11).
Coverage Gap	employees with zero attendance in period	New. Replaces the manual Calc sheet.
Months Since Last Training	per employee, from latest attendance date	New.

REQUIRES STAKEHOLDER SIGN-OFF BEFORE LAUNCH

Correcting Participation Rate and NPS makes the headline numbers look worse. Participation falls from 66.7% to **60.4%**, and per-trainer NPS will shift once name variants are merged. This must be walked through with L&D before go-live, supported by the reconciliation in section 14, and not discovered afterwards.

10. Functional requirements

Priority key: **MUST** ships in version 1. **SHOULD** ships in version 1 if capacity allows. **COULD** is a candidate for a later release.

10.1 Module A: Integration and synchronisation

ID	Requirement	Priority
FR-A01	Sync Learning Programs from the CRM: ID, title, subtitle, target, type, track, capacity, status, start date, description.	MUST
FR-A02	Sync sessions per program, including date, start time and end time.	MUST
FR-A03	Sync enrollments per program, including the enrolled employee and enrollment status.	MUST
FR-A04	Sync attendance records per session, as captured by CRM QR scan.	MUST
FR-A05	Sync feedback responses, including all five scored questions and the free-text comment.	MUST
FR-A06	Sync employee master from the HRIS: code, email, name, sector, department, position, job level, payscale cluster, hire date, status.	MUST
FR-A07	Run incremental sync every 30 minutes, filtered on source updated_at.	MUST
FR-A08	Run a nightly full reconciliation sync that detects records deleted or silently edited at source.	MUST
FR-A09	Persist every source response unmodified in the raw layer, and support full replay of transforms from it.	MUST
FR-A10	Make all sync operations idempotent. Re-running any window must not duplicate or corrupt records.	MUST
FR-A11	Log every sync run with source, window, record counts, duration and outcome; alert the team on failure.	MUST
FR-A12	Allow an operator to trigger an on-demand sync for a chosen source and date range.	SHOULD
FR-A13	Retry transient CRM and HRIS failures with exponential backoff, and degrade to serving last-known-good data rather than erroring.	MUST

10.2 Module B: Data model and conformance

ID	Requirement	Priority
FR-B01	Maintain the dimensional model in section 8, with facts held at three separate grains.	MUST
FR-B02	Maintain dim_employee as SCD Type 2, so headcount can be evaluated as of any historical date.	MUST
FR-B03	Derive session duration from start and end times; flag any session where duration cannot be derived.	MUST
FR-B04	Resolve attendee identity by employee code, then email, then normalised name, in that order.	MUST
FR-B05	Quarantine any record whose identity cannot be resolved, and exclude it from metrics until resolved. Never drop it silently.	MUST
FR-B06	Conform trainer names through an alias table so variants resolve to a single trainer.	MUST
FR-B07	Normalise all categorical values on ingest (trim whitespace, standardise case) and reject unrecognised values into the exception queue.	MUST
FR-B08	Key all program facts on CRM program ID; never on program title.	MUST
FR-B09	Backfill February to August 2026 history from the existing workbook where CRM records are unavailable, tagged with its source.	MUST

10.3 Module C: Enrichment administration

ID	Requirement	Priority
FR-C01	Provide a screen listing all programs with missing enrichment, so L&D can assign customised department and trainer.	MUST
FR-C02	Support assigning the same value to multiple programs in one action.	SHOULD
FR-C03	Record who changed each enrichment value and when, and retain prior values.	MUST
FR-C04	Where the CRM later supplies a field held as an override, prefer the CRM value and mark the override redundant.	SHOULD
FR-C05	Never write enrichment values back to the CRM.	MUST

10.4 Module D: Dashboard and analysis

ID	Requirement	Priority
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FR-D01	Present a KPI summary covering programs, training days, participants, training hours, learner hours, participation rate, NPS and the four quality dimensions.	MUST
FR-D02	Apply global filters (date range, sector, department, program type, target, trainer, program) consistently across every view.	MUST
FR-D03	Allow any KPI or chart element to drill through to the underlying records that produced it.	MUST
FR-D04	Show every KPI as a monthly trend as well as a period total.	MUST
FR-D05	Provide a program scorecard: sessions, capacity, enrollments, fill rate, attendees, no-show rate, hours, NPS, quality scores and response rate.	MUST
FR-D06	Provide a trainer scorecard: programs and sessions delivered, learners reached, correctly weighted NPS, four quality dimensions, response count and trend.	MUST
FR-D07	Provide a coverage view: participation rate by sector, department and job level, the list of employees with zero training in the period, and months since last training.	MUST
FR-D08	Provide a funnel view: enrolled to attended to evaluated, per program and in total, with no-show and response rates.	MUST
FR-D09	Provide a learner profile: an individual's programs attended, total learning hours, and evaluations submitted.	MUST
FR-D10	Provide a top-learners ranking by learning hours, derived automatically, replacing the hand-typed sheet.	MUST
FR-D11	Display data freshness (last successful sync time) on every screen.	MUST
FR-D12	Warn on any view whose figures are affected by open data-quality exceptions, stating how many records are excluded.	MUST
FR-D13	Allow a user to save a filter combination as a named view and return to it.	SHOULD
FR-D14	Allow two periods to be compared side by side.	COULD

10.5 Module E: Reporting and export

ID	Requirement	Priority
FR-E01	Export any on-screen view to CSV or XLSX, honouring the filters currently applied.	MUST
FR-E02	Generate a formatted monthly XLSX report reproducing the existing dashboard layout, so recipients need no retraining.	MUST
FR-E03	Generate a PDF report pack for HR leadership.	SHOULD

FR-E04	Stamp every export with generation timestamp, applied filters, data-freshness time, and the count of excluded records.	MUST
FR-E05	Schedule reports to generate and email to a recipient list on a defined cadence, with no human step.	MUST
FR-E06	Retain generated reports and allow prior editions to be retrieved.	SHOULD
FR-E07	Compute all export figures from the same metric layer as the dashboard, never a parallel implementation.	MUST

10.6 Module F: Data quality

ID	Requirement	Priority
FR-F01	Provide an exception queue listing every open data-quality issue with its type, affected records and suggested resolution.	MUST
FR-F02	Detect at minimum: unresolvable attendee identity, program missing a trainer, customised program missing a department, session missing a duration, attendance without a matching enrollment, evaluation for a program with no attendance, and duplicate attendance for the same employee and session.	MUST
FR-F03	Allow an operator to resolve an exception by mapping to an existing entity, creating an override, or dismissing it with a recorded reason.	MUST
FR-F04	Show a completeness indicator per reporting period: percentage of records fully resolved.	MUST
FR-F05	Never exclude a record from reporting without registering an exception for it.	MUST

11. Non-functional requirements

ID	Area	Requirement
NFR-01	Performance	Any dashboard view returns within 2 seconds at the 95th percentile, on five years of projected data (approximately 15,000 attendance records).
NFR-02	Freshness	Dashboard data is never more than 30 minutes behind the CRM under normal operation.
NFR-03	Availability	99% during business hours. A CRM or HRIS outage degrades freshness, never availability: last-known-good data continues to serve.
NFR-04	Authentication	Company SSO. No local passwords.
NFR-05	Authorisation	Single L&D role in version 1. The model must accommodate row-level scoping later without restructuring.
NFR-06	Data protection	The platform holds employee PII: names, emails, phone numbers, national IDs, pay grades. National ID and payscale data must not be ingested unless a stated metric requires them. Encryption in transit and at rest.
NFR-07	Auditability	Every published figure traces to its source records. Enrichment changes and exception resolutions are attributed and retained.
NFR-08	Observability	Structured JSON logging. Sync health, freshness lag and exception volume monitored with alerting.
NFR-09	Scalability	Services are stateless and horizontally scalable; state is held in PostgreSQL and Redis.
NFR-10	Correctness	Every metric in section 9 has automated tests asserting its value against a fixed reference dataset. A metric change that alters a published figure fails the build until reviewed.
NFR-11	Configuration	All environment-specific values and credentials via environment variables. No secrets in source.
NFR-12	Browser support	Current Chrome, Edge, Firefox and Safari. Desktop-first; readable on tablet.
NFR-13	Accessibility	WCAG 2.1 AA for contrast, keyboard navigation and focus visibility.
NFR-14	API versioning	All endpoints under a /v1/ prefix.

DELIBERATE CONSTRAINT ON INFRASTRUCTURE

Current volume is 55 programs, 212 employees and roughly 400 attendance records, growing by an estimated 2,000 rows per year. A single PostgreSQL instance will serve this comfortably for a decade. The team should **not** introduce a data warehouse, columnar store, dbt or a streaming pipeline. Complexity here buys nothing and costs maintenance forever.

12. Reports and export

Report	Contents	Format	Cadence
Monthly L&D Report	Twelve headline KPIs, monthly trend, program and trainer breakdowns, public versus customised split, internal versus external split.	XLSX + PDF	Scheduled monthly
Coverage Report	Participation by sector, department and job level; employees with zero training; months since last training.	XLSX	On demand and quarterly
Program Scorecard	Full delivery and quality profile for one program, including all free-text comments.	PDF	On demand
Trainer Scorecard	Delivery volume and quality performance for one trainer over a chosen period.	PDF	On demand
Attendance Extract	Row-level attendance with all dimensional attributes joined. The analytical equivalent of today's master sheet.	CSV / XLSX	On demand
Exception Report	All open data-quality issues by type and age.	XLSX	On demand and weekly

The Monthly L&D Report deliberately mirrors the current dashboard's layout and vocabulary. Recipients outside L&D should receive something they already recognise; the change is that nobody assembles it.

13. Data quality and exception handling

The governing rule: **a record is either counted or registered as an exception, never neither.** The workbook's most dangerous behaviour is silent loss, where 38 attendees with no employee code simply vanish from sector reporting with no indication anything is missing.

Rule	Trigger	Handling
IDENTITY_UNRESOLVED	Attendee matches no employee by code, email or name	Quarantine. Excluded from metrics, surfaced for manual mapping.
TRAINER_MISSING	Program has no trainer from CRM or override	Counted in volume metrics, excluded from trainer scorecards.
CUSTOMISED_DEPT_MISSING	Program target is Department but no department assigned	Counted in totals, excluded from department breakdowns.
DURATION_UNDERIVABLE	Session missing start or end time	Excluded from both hours metrics, flagged for correction.
ATTENDANCE_NO_ENROLLMENT	Attendance exists with no matching enrollment	Counted in attendance, excluded from funnel. Likely a walk-in.
EVALUATION_NO_ATTENDANCE	Feedback submitted by someone with no attendance record	Flagged. Indicates an attendance capture failure.
DUPLICATE_ATTENDANCE	Same employee recorded twice at the same CRM session . Grain must be the CRM session ID, never the workbook # column: on (employee, date, program) the workbook shows 7 duplicates, but on (employee, #) it shows 90 – see P-12.	Deduplicated to one record; the duplicate is logged.
CAPACITY_EXCEEDED	Enrollments exceed program capacity	Reported as-is, flagged for review.

14. Migration and reconciliation

Before the platform becomes the system of record for reporting, the team must produce a **reconciliation statement**: every KPI, side by side, workbook value against platform value, with a stated cause for each difference.

KPI	Workbook	Expected	Cause of difference
Participation Rate	66.7%	60.4%	Denominator corrected from 192 to actual headcount of 212 (P-01)
Total Programs	24	To be confirmed; higher	Title-keyed merges separated (P-02); previously unreported programs included (P-08)
NPS (overall)	92.7%	84.4%	Hidden Completion Time filter removed; all 77 responses included (P-03)
Logistics Effectiveness	96.4%	92.2%	Same filter removed (P-03)
Knowledge Relevance	100.0%	97.4%	Same filter removed (P-03)
Activity Effectiveness	98.2%	97.4%	Same filter removed (P-03)
Facilitator Performance	100.0%	100.0%	Unchanged once the filter is removed (P-03)
Per-trainer NPS	varies	shifts	Name variants merged (P-04); response scope widened (P-03)
Sector breakdowns	varies	shifts	Whitespace variants merged (P-05); the 38 unjoinable rows (12 people) resolved or quarantined (P-07)
Training Days	40	To be confirmed	Distinct CRM sessions. Depends on resolving P-12 first
Training Hours Delivered	130.5	To be confirmed	Derived from session times rather than hand-entered; scope widened to all CRM programs
Total Participants	128	To be confirmed;	Programs absent from the workbook now included

		higher	
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The reconciliation is a **launch gate**, not a post-launch artefact. L&D must accept each restated figure before the platform is published. Run both systems in parallel for one full reporting cycle.

15. Assumptions, dependencies and risks

15.1 Assumptions

- The CRM exposes, or can expose, REST endpoints for programs, sessions, enrollments, attendance and feedback.
- The HRIS is integrable by API or read replica and holds the full active roster.
- CRM attendance capture via QR is complete and reliable for the periods reported.
- Feedback is captured in the CRM, as implied by its Export Attendance & Feedback action, rather than in a separate survey tool.
- L&D will maintain enrichment values promptly. Stale enrichment degrades breakdowns, not totals.

15.2 Risks

ID	Risk	Impact	Mitigation
R-01	CRM API does not exist, is incomplete, or the CRM team cannot prioritise it	High	Confirm the integration surface in week 1. Fall back to the Export Attendance & Feedback file ingest as an interim path.
R-02	Trainer is absent from the CRM entirely, making all trainer analysis dependent on manual entry	Medium	Verify in week 1. If absent, scope the enrichment screen for per-program trainer assignment and request the field from the CRM roadmap.
R-03	Restated KPIs are rejected by stakeholders because the numbers look worse	High	Present the reconciliation early, in week 3, alongside the evidence. Frame as correcting a known defect, not as a performance decline.
R-04	Historical CRM data is thinner than the workbook for February to August 2026	Medium	Backfill from the workbook, tagged by source. Accept lower fidelity before cutover and mark it in the interface.
R-05	HRIS lacks employee history, preventing correct as-of headcount	Medium	Begin capturing SCD Type 2 snapshots from day one; use current-state headcount for backfilled periods and label it.
R-06	Identity resolution fails at scale, for contractors and new hires with no employee code	Medium	Exception queue plus manual mapping. Report unresolved volume as a completeness measure rather than hiding it.
R-07	Employee PII expands the system's	Medium	Ingest only fields a stated metric requires.

R-08

data-protection obligations		Exclude national ID and payscale unless justified. Review with HR before build.
The team over-engineers infrastructure for a very small dataset	Medium	The constraint under section 11 is explicit. PostgreSQL only; no warehouse tooling.

16. Open questions

Each of these must be answered before the affected requirement can be built. Owner and target date to be assigned at kick-off.

ID	Question	Blocks	Owner
Q-01	Does the CRM hold the trainer for each program or session, and is it exposed by the API?	FR-A01, FR-C01, FR-D06	
Q-02	For programs targeted at a Department, does the CRM record which department?	FR-C01	
Q-03	Is feedback stored in the CRM, or still collected in a separate survey tool?	FR-A05	
Q-04	What does the CRM Enrollments tab record: per program or per session?	FR-A03, FR-D08	
Q-05	Which HRIS is in use, and does it retain employee history or only current state?	FR-A06, FR-B02	
Q-06	How are external attendees and new hires without an employee code to be treated in participation rate?	FR-B04	
Q-07	Does an annual training plan exist in any structured form? If so, plan-versus-actual returns to scope.	Section 5	
Q-08	Is a LinkedIn Learning data export available, and at what grain?	Section 5	
Q-09	Which workbook is referenced by the broken [1]Sheet1 link, and is anything in it still required?	P-10, FR-B09	
Q-11	Why does the Completion Time filter exist on the evaluation pivot, and was excluding those 22 responses deliberate?	P-03, Section 14	
Q-12	Does the CRM expose a stable session identifier per attendance record? Without one, historical Training Days cannot be restated.	P-12, FR-A02, FR-B09	
Q-10	Should the 30 CRM programs absent from the workbook be reported historically, or only from cutover?	FR-B09, Section 14	

17. Delivery plan

Timing	Phase	Contents and exit criterion
Weeks 1-2	Foundation and integration	Resolve Q-01 to Q-05. Build CRM and HRIS clients, the raw landing layer, and sync orchestration with logging and alerting. Exit: both sources sync on schedule, and raw data is replayable.
Weeks 3-4	Model, metrics and reconciliation	Star schema, identity resolution, conformance, and the metric layer with its full test suite. Produce the section 14 reconciliation and walk L&D through every restated figure. Exit: every KPI computes correctly and each difference from the workbook is accepted.
Weeks 5-6	Core dashboard	API, React shell, KPI summary, global filters, monthly trends, drill-through, and the enrichment administration screen. Exit: L&D can reproduce their current dashboard on the platform.
Weeks 7-8	Analysis views and export	Program and trainer scorecards, coverage view, funnel view, learner profile, top learners. CSV and XLSX export, and the formatted monthly report. Exit: every analysis in section 5 is live and exportable.
Week 9	Data quality and automation	Exception queue with the full rule set, completeness indicators, scheduled report generation and email delivery. Exit: the monthly report sends with no human step.
Week 10	Parallel run and handover	Run alongside the workbook for one full cycle. User acceptance testing, performance validation, documentation, training. Exit: sign-off and workbook retirement.

18. Acceptance criteria

The platform is accepted when all of the following hold:

1. Every CRM Learning Program appears in the dashboard, and the count matches the CRM exactly.
2. Every KPI in section 9 computes to a value L&D has signed off in the section 14 reconciliation.
3. Producing the monthly report requires no manual data assembly, only enrichment review and exception clearing, in under 30 minutes.
4. The scheduled monthly report generates and delivers by email with no human intervention, for one full cycle.
5. Any KPI can be drilled through to the source records that produced it.
6. No record is excluded from reporting without a corresponding exception being registered.
7. Every view returns within 2 seconds at the 95th percentile.
8. The automated metric test suite passes against the reference dataset.
9. Coverage, funnel and scorecard views are live and exportable.
10. A full parallel run against the workbook has completed with all differences explained.